MLP interventions versus the unchanged Adam baseline

All interventions freeze first-linear weights + biases at 448; I2 onward additionally change LR at 500. Baseline remains unchanged.

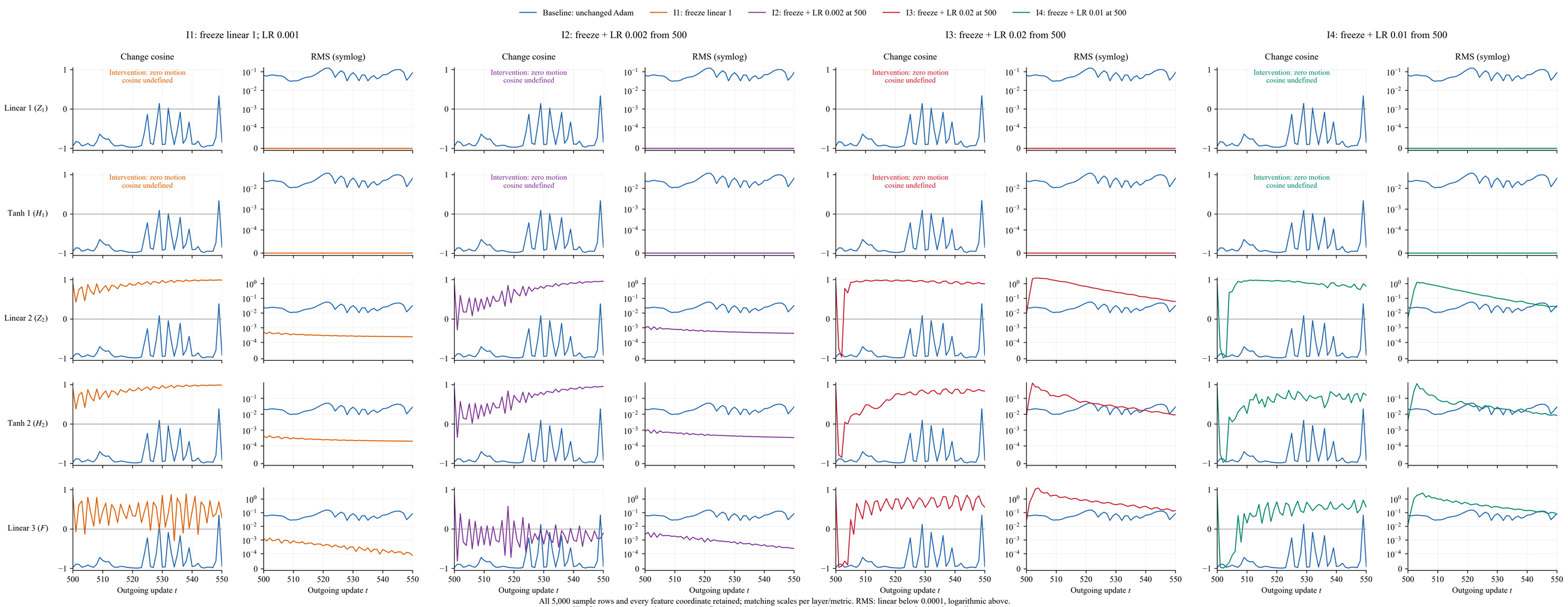
Baseline: unchanged Adam · I1: freeze linear 1 · I2: freeze + LR 0.002 at 500 · I3: freeze + LR 0.02 at 500 · I4: freeze + LR 0.01 at 500

I1: freeze linear 1; LR 0.001 · I2: freeze + LR 0.002 from 500 · I3: freeze + LR 0.02 from 500 · I4: freeze + LR 0.01 from 500

(a) Training loss (log)
(b) $\eta\hat{\kappa}_t$ (symlog)
(c) Gradient cosine
(d) Full $\|g_t\|$ (log)
(e) Later-layer $\|g_t\|$ (log)
(f) Output-change cosine
(g) Output RMS (log)
(h) LR of later layers

Completed optimizer updates t

$\eta\hat{\kappa}_t = 1 - \langle g_t, g_{t-1}\rangle / \|g_{t-1}\|^2$: projected-gradient diagnostic, not an Adam curvature threshold.
Row (c): solid = all 656,810 parameters; dashed = the same 42,210 later-layer parameters used in row (e).
Same 5,000 examples and axis scales. Output changes are outgoing. Shading: 500-550; dotted line: LR change. Log scales display the full dynamic range.

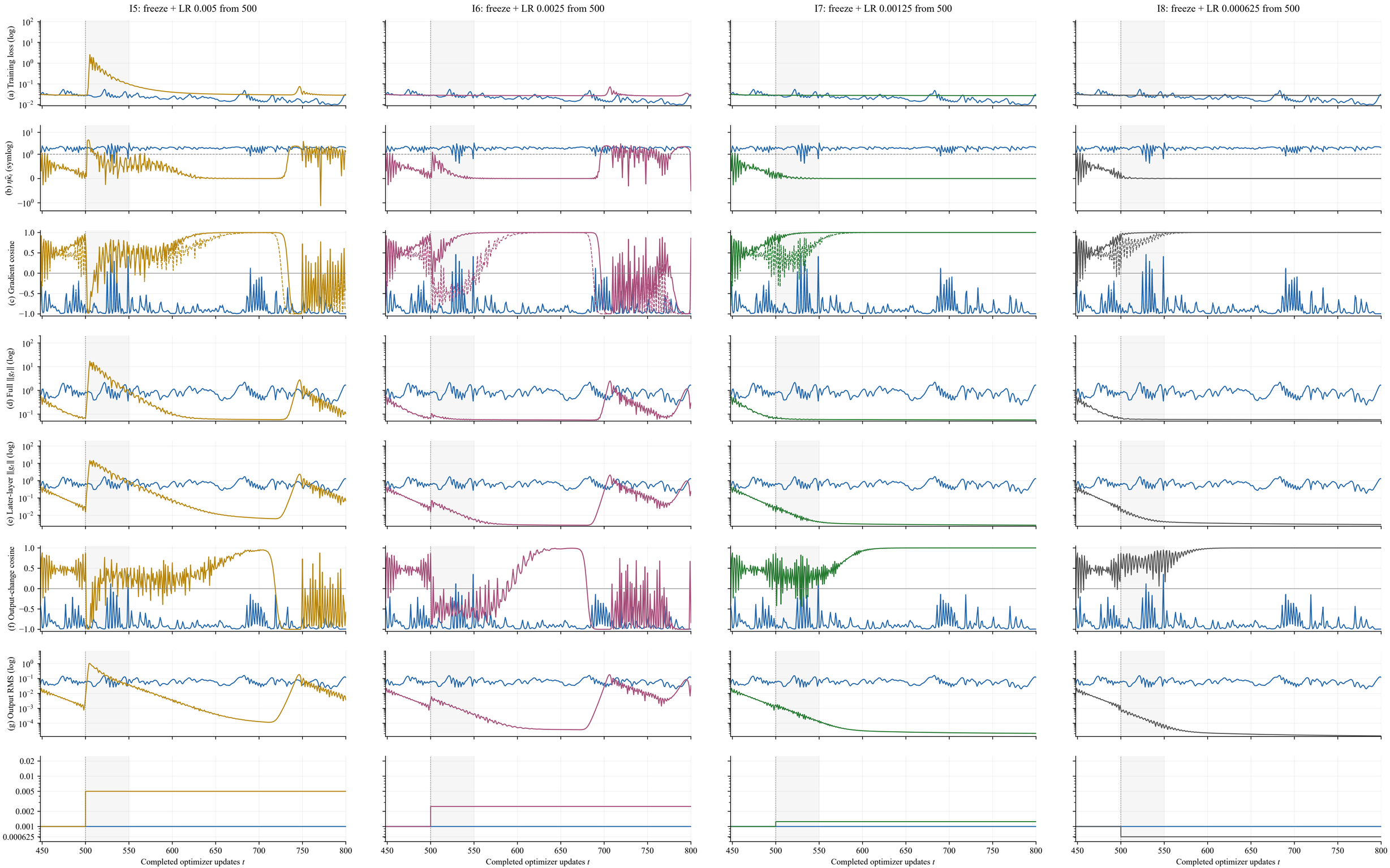


All 5,000 sample rows and every feature coordinate retained; matching scales per layer/metric. RMS: linear below 0.0001, logarithmic above.
The first two representations are exactly fixed only in the intervention. Negative cosine does not alone imply a loss spike.

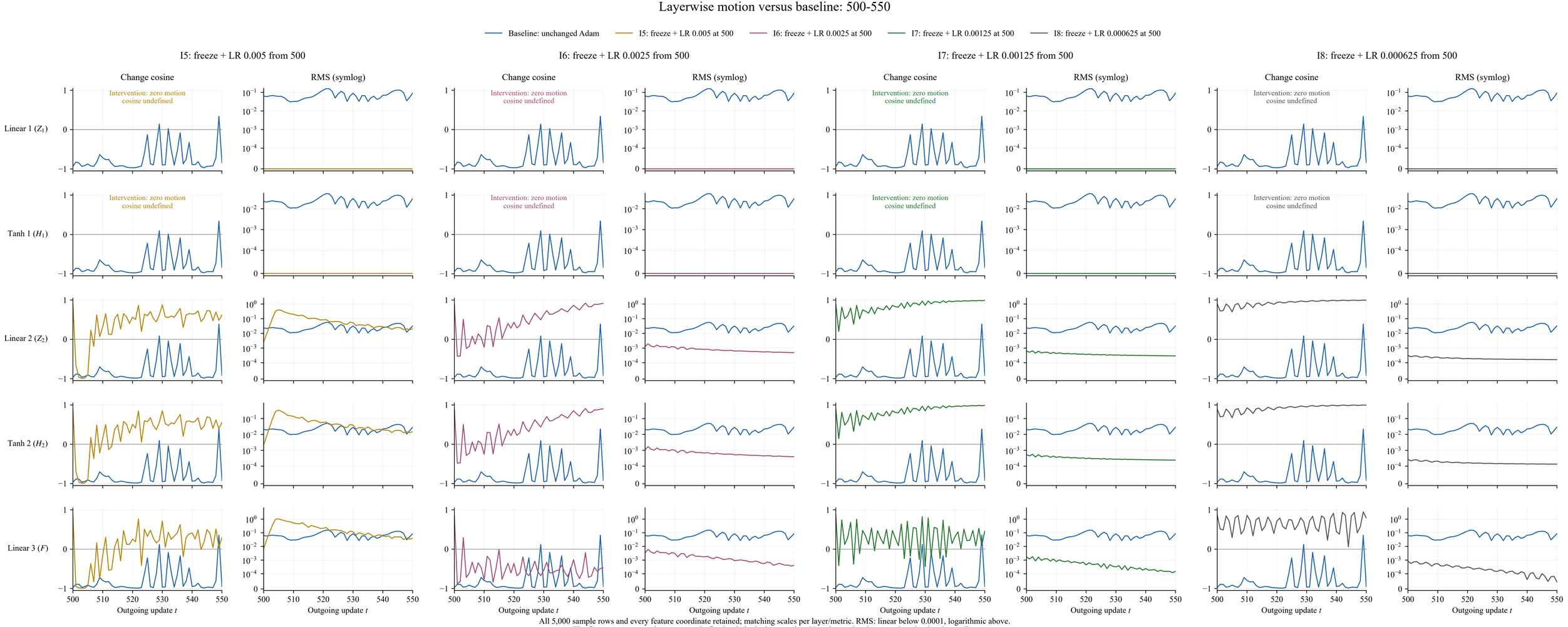
MLP interventions versus the unchanged Adam baseline

All interventions freeze first-linear weights + biases at 448; I2 onward additionally change LR at 500. Baseline remains unchanged.

Baseline: unchanged Adam · I5: freeze + LR 0.005 at 500 · I6: freeze + LR 0.0025 at 500 · I7: freeze + LR 0.00125 at 500 · I8: freeze + LR 0.000625 at 500



$\eta\hat{\kappa}_t = 1 - \langle g_t, g_{t-1}\rangle / \|g_{t-1}\|^2$: projected-gradient diagnostic, not an Adam curvature threshold.
Row (c): solid = all 656,810 parameters; dashed = the same 42,210 later-layer parameters used in row (e).
Same 5,000 examples and axis scales. Output changes are outgoing. Shading: 500-550; dotted line: LR change. Log scales display the full dynamic range.


Layerwise motion versus baseline: 500-550
Baseline: unchanged Adam
I5: freeze + LR 0.005 at 500
I6: freeze + LR 0.0025 at 500
I7: freeze + LR 0.00125 at 500
I8: freeze + LR 0.000625 at 500
I5: freeze + LR 0.005 from 500
I6: freeze + LR 0.0025 from 500
I7: freeze + LR 0.00125 from 500
I8: freeze + LR 0.000625 from 500
Change cosine
RMS (symlog)
Intervention: zero motion
cosine undefined
Linear 1 (Z_1)
Tanh 1 (H_1)
Linear 2 (Z_2)
Tanh 2 (H_2)
Linear 3 (F)
Outgoing update t
All 5,000 sample rows and every feature coordinate retained; matching scales per layer/metric. RMS: linear below 0.0001, logarithmic above.
The first two representations are exactly fixed only in the intervention. Negative cosine does not alone imply a loss spike.